Analyzing Felleman & Van Essen paper (Distributed Hierarchial processing in the primate Cerebral Cortex)

- The Macaque visual cortex contains 32 distinct cortical areas connected by 305 identified pathways. They follow a structural rule that lets you infer a hierarchy of ~10 processing levels from retina to the highest association cortex.

Cortical Layers

1 Few cell bodies (mostly axons & Dendrites) :-
Feedback target - receives top-down signal

2/3 Small pyramidal neurons :
Feedforward output - Send signals Up the hierarchy

4. Spiny Stellate cells (in V1)
Feedforward Input - receives signals from lower areas

5. Large pyramidal neurons -
Output to subcortical structures (superior Colliculus, pons)

6. Polymorphic Neurons -
Feedback to thalamus (LGN)

* CNN : layer 4 - input tensor to a conv block, layer 2/3 output feature passed to the next block.

the hierarchy role

Type	Origin Layers	Termination	What it means
Feedforward (ascending)	Superficial (2/3) and Sometimes 5	Mainly layer 4	Sends processed info up to a higher area.
Feedback (descending)	Deep (5/6) and Sometimes 2/3	layers 1, 2/3, 5/6 - Skips layer 4	Sends top-down modulation down to a lower area.
Lateral	Mixed layers	Columnar - all layers roughly equally	Connects area to the same hierarchical level.

$A \xrightarrow{\text{feedforward}} B$, A is lower in hierarchy

$A \xrightarrow{\text{Feedback}} B$, A is higher in hierarchy

$A = B$ (connection lateral) $\rightarrow$ Same level.

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* CNNs are only feedforward - no feedback connections. Recurrent Networks and top-down attention mechanisms (transformers) partially address this but biological feedbacks are far reacher.

4) <u>Visual Stream Hierarchy</u>

32 areas are organized into two major processing streams that diverge after V1 and V2

Ventral Stream ("What" pathway):

- V1 → V2 → V4 → PIT → CIT → AIT (Inferotemporal cortex)
- Processes object identity, shape, color, texture, face
- Damage to IT Cortex → Prosopagnosia (inability to recognize faces)

Dorsal Stream ("Where / How"):

V1 → V2 → V3 → MT/V5 → MST → posterior parietal cortex

- Processes spatial location, motion, depth and action guidance
- Damage to MT → akinetopsia (inability to perceive motion)